

Power and Cooling -- Guided Notes ANSWER KEY

Unit 2.4 | CompTIA Network+ N10-009 Obj. Core 1 3.5 | N10-008

For instructor use / distribute after students complete the notes.

Question 1

The ATX PSU provides three main DC voltage rails. The _____ V rail powers the CPU and GPU and is the most heavily loaded rail. The _____ V rail powers older drives and some motherboard logic. The _____ V rail powers RAM and low-voltage logic. The wattage rating of the PSU represents its maximum _____ output.

Answer: +12V; +5V; +3.3V; continuous (sustained) power.

Question 2

The 80 PLUS certification rates PSU _____. An 80 PLUS Gold PSU achieves approximately _____ % efficiency at 50% load, meaning _____ % of the power drawn from the wall is wasted as heat. A more efficient PSU produces less _____ and reduces electricity costs.

Answer: Efficiency (energy efficiency); 90%; 10%; heat.

Question 3

When selecting a PSU wattage, technicians add up the TDP of all components and add _____ to _____ % as a headroom buffer. This prevents the PSU from running near _____ % capacity, which reduces efficiency and component lifespan. A system with a 300W total TDP should use a PSU rated at least _____ W.

Answer: 20% to 30%; 100%; 360-390W (300W + 20-30%).

Question 4

The EPS _____ -pin connector (sometimes called the CPU power connector) delivers +12V power directly to the _____ voltage regulator module on the motherboard. Without this connector plugged in, high-end CPUs may _____ POST or operate at reduced power.

Answer: 8-pin; CPU (motherboard VRM); fail to.

Question 5

Thermal paste (also called thermal compound) is applied between the CPU and heatsink to fill _____ air gaps in the metal surfaces. It improves _____ transfer from the CPU to the heatsink. The recommended application amount is approximately _____ -sized -- too much can _____ onto the motherboard and cause shorts.

Answer: Microscopic (tiny); heat; pea-sized; spill (spread).

Question 6

An AIO (All-In-One) liquid cooler is a _____ cooling system. It consists of a _____ block that mounts on the CPU, connected by tubes to a _____ where heat is dissipated. AIO coolers generally provide _____ thermal performance than air coolers for high-TDP processors, but require more

_____ mounting space.

Answer: Closed-loop; pump and cold plate (water block); radiator; better (superior); case (radiator).

Question 7

Case fans are labeled with a direction indicator. The _____ side of the fan is the exhaust side. For optimal airflow, intake fans should be mounted at the _____ and front, while exhaust fans go at the _____ and top. A system with more _____ fans than exhaust fans creates positive pressure, which reduces dust buildup.

Answer: Label (sticker) side; bottom; rear; intake.

Question 8 -- Real World Application

REAL WORLD: A student builds a PC and notices that after 10 minutes of gaming, the system stutters, frame rates drop sharply, and then performance returns to normal after a minute. No BSOD occurs. Explain the most likely cause, identify the technical term for this behavior, and describe two corrective actions the technician should take.

Answer: The most likely cause is thermal throttling. When the CPU or GPU reaches its maximum safe operating temperature, it reduces its clock speed to lower heat production -- this causes the performance drop. After a minute of reduced load, temperatures drop and performance recovers. Technical term: thermal throttling (also called dynamic thermal management). Corrective actions: (1) Verify thermal paste is applied correctly -- reapply if the CPU temperature exceeds 90-95 degrees C under load. (2) Check case airflow -- add intake fans, clear cable obstructions, and ensure the exhaust path is unblocked. Also verify the CPU cooler is properly seated with even contact pressure.